

Lab Task 2: Word Analysis & Tokenization

AIM:

To perform word analysis by tokenizing text using:

- the Transformers library
- stopwords as delimiters using NLTK and spaCy

CODE:

```
# !pip install transformers nltk spacy

from transformers import AutoTokenizer

import nltk

import spacy

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

# Download NLTK data

nltk.download('punkt')

nltk.download('punkt_tab')

nltk.download('stopwords')

# Load spaCy model

# python -m spacy download en_core_web_sm

nlp = spacy.load("en_core_web_sm")

# Sample text

text = "Natural Language Processing is an exciting field of Artificial Intelligence."

# ♠ Part A: Transformers Tokenization (BERT)

tokenizer = AutoTokenizer.from_pretrained("bert-base-uncased")

tokens_transformers = tokenizer.tokenize(text)

print("Transformers Tokens:")

print(tokens_transformers)

# ♠ Part B: NLTK Tokenization + Stopword Removal

tokens = word_tokenize(text)

stop_words = set(stopwords.words('english'))

filtered_tokens_nltk = [word for word in tokens if word.lower() not in stop_words]
```

```
print("\nNLTK Tokens (Stopwords Removed):")
print(filtered_tokens_nltk)

# ♠ Part C: spaCy Tokenization + Stopword Removal

doc = nlp(text)

filtered_tokens_spacy = [token.text for token in doc if not token.is_stop]

print("\nspaCy Tokens (Stopwords Removed):")

print(filtered_tokens_spacy)
```

Output:

Transformers Tokens:

```
['natural', 'language', 'processing', 'is', 'an', 'exciting',
 'field', 'of', 'artificial', 'intelligence', '.']
```

NLTK Tokens (Stopwords Removed):

```
['Natural', 'Language', 'Processing', 'exciting', 'field',
 'Artificial', 'Intelligence', '.']
```

spaCy Tokens (Stopwords Removed):

```
['Natural', 'Language', 'Processing', 'exciting', 'field',
 'Artificial', 'Intelligence', '.']
```